

Data Science & AI for Economists

Lecture 4: Introduction to Nighttime light data and GIS applications

Zhaopeng Qu

Business School, Nanjing University

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Roadmap

Last Thursday and Today's Agenda

- Intro to GIS
 - What is GIS?
 - Why is GIS useful for Economists?
 - GIS Data Types and Coordinate Systems
- Basic GIS analysis in R:
 - Intro to R packages: sf, raster, tmap, leaflet
 - Using R to open map data
- Nightlights Data in Economics
 - What and Why NTL?
 - Some Applications of NTL in Economics
 - How to get NTL data?
 - How to process NTL data?
- Using NTL data to practice GIS Analysis in R
 - Using package blackmarbleR to download NTL data
 - Using package raster and sf to process NTL data

Introduction to GIS

Introduction to Nightlights Data

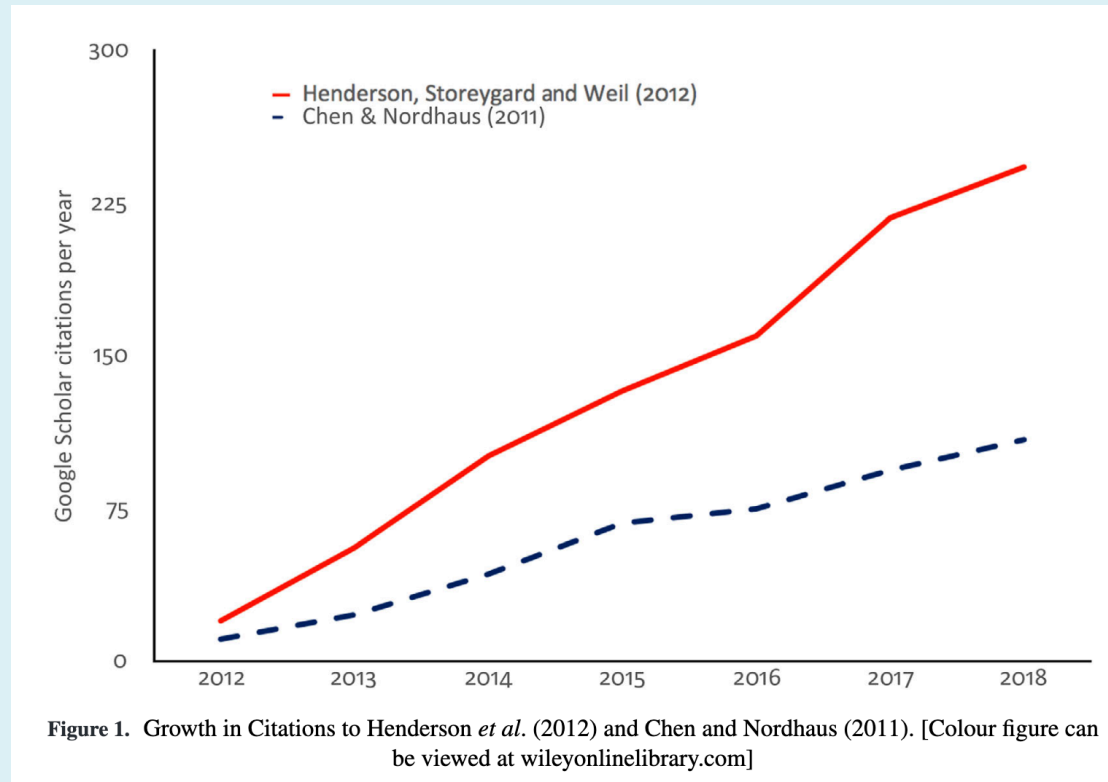
Introduction to Nightlights Data

- Nightlights is a specific image data by **remote sensing** in satellites that captures the nighttime light of the earth.



Nightlights Data in Economics

- Two pioneering applications of NTL in Economics are increasingly cited in the literature:
 - Henderson and Storeygard et al (2012) and Chen and Nordhaus (2011)



Nightlights Data in Economics

- The most-cited papers using NTL data are also not only in remote sensing field but also in economics field.

Table 1. Top 10 most-cited articles using NTL data.

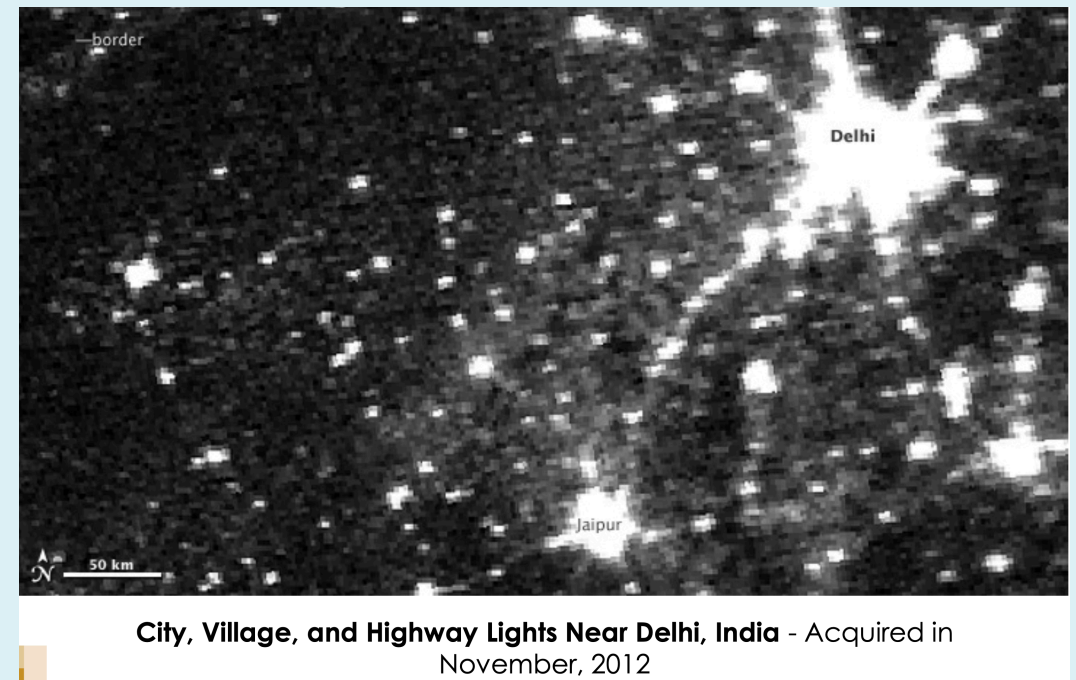
Rank	Cites	Title	Journal	Year	Prior Top10	Economists
1	3229	Measuring economic growth from outer space	<i>AmerEconRev</i>	2012	No	Yes
2	1628	The new world atlas of artificial night sky brightness	<i>SciAdv</i>	2016	No	No
3	1559	Pre-colonial ethnic institutions and contemporary African development	<i>Econometrica</i>	2013	No	Yes
4	1407	Using luminosity data as a proxy for economic statistics	<i>PNAS</i>	2011	No	Yes
5	1147	Relation between satellite observed visible-near infrared emissions, population, economic activity and electric power consumption	<i>IntJRemSens</i>	1997	#6	No
6	1090	The first world atlas of artificial night sky brightness	<i>NotRoyAstSoc</i>	2001	#5	No
7	1124	Regional favoritism	<i>QuartJEcon</i>	2014	No	Yes
8	984	Mapping city lights with nighttime data from the DMSP operational linescan system	<i>PhotoEngRS</i>	1997	#2	No
9	891	VIIRS night-time lights	<i>IntJRemSens</i>	2017	No	No
10	861	Ethnic inequality	<i>JPolitEcon</i>	2016	No	Yes

Notes: The citations are as of 5 November 2024, in *Google Scholar*. The prior top 10 refers to papers listed in Table 5 of [11]. The rank for each article provides the number in the reference list where bibliographic details can be found.

Introduction to Nightlights Data

Previous generation of NTL Products

- DMSP/OLS (Defense Meteorological Satellite Program/Operational Linescan System)
 - Operated by the U.S. Department of Defense.
 - from 1992 to 2013
 - Monthly product
 - lack of resolution



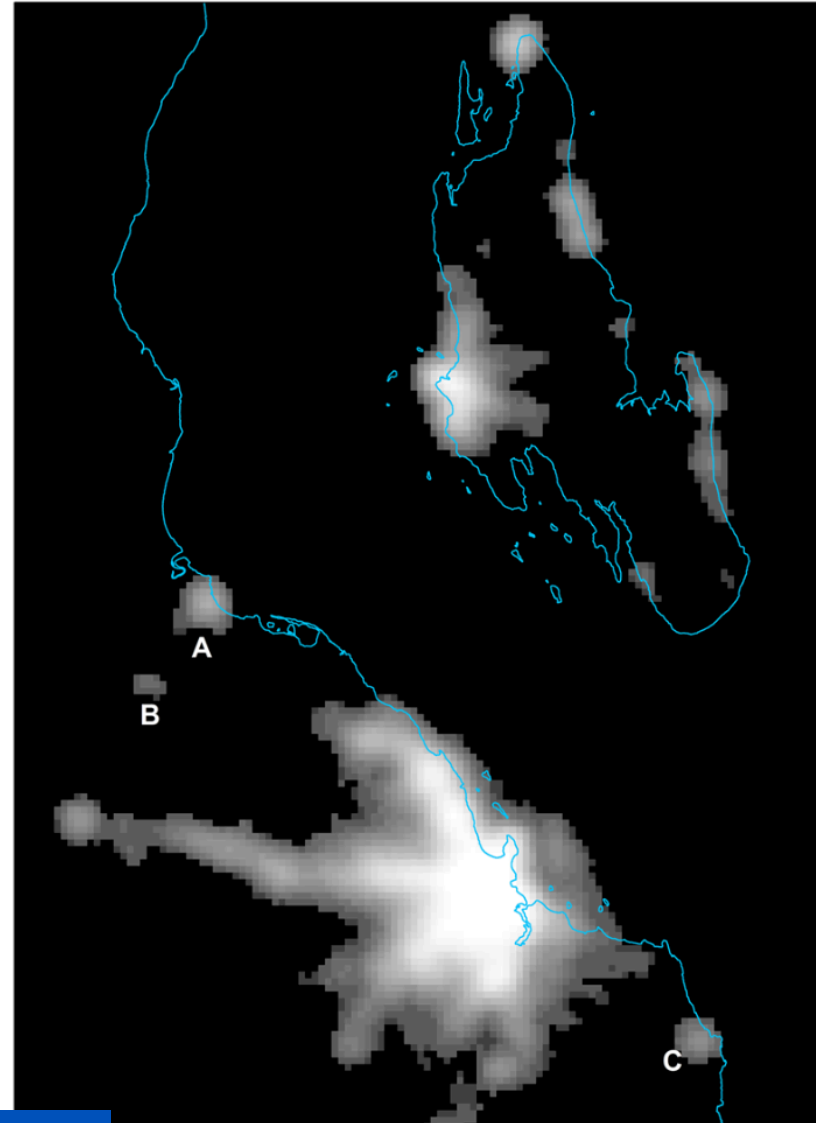
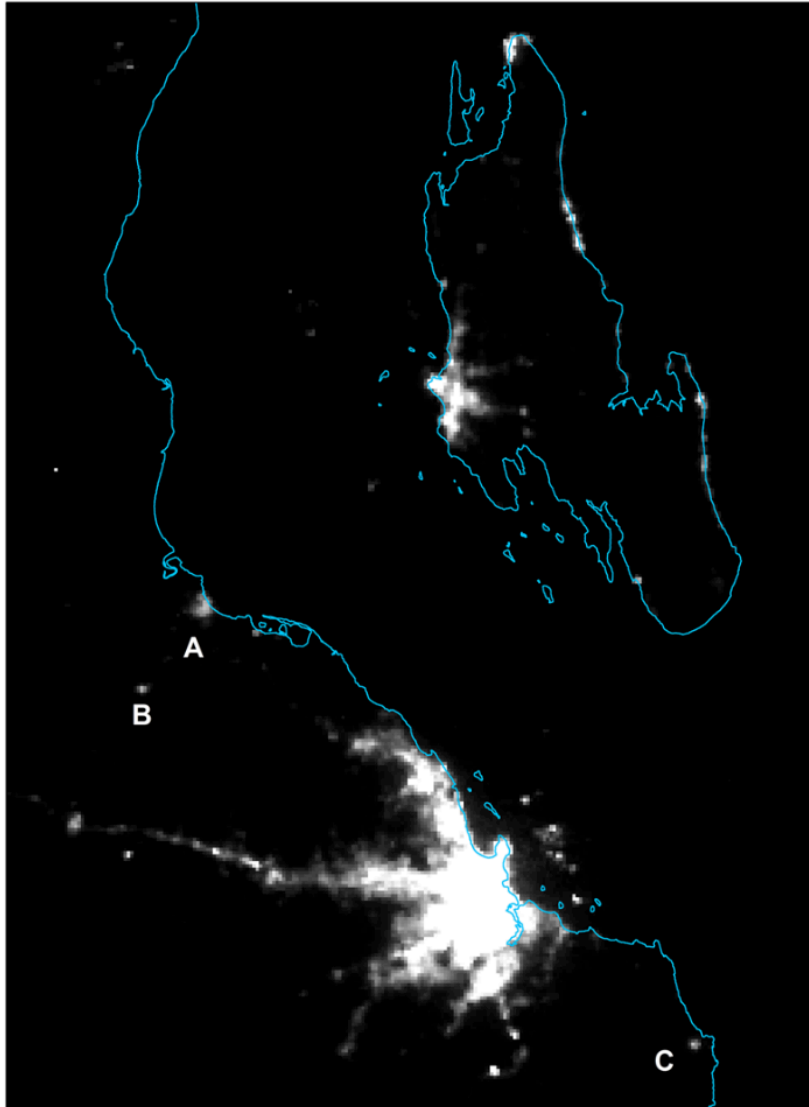
Introduction to Nightlights Data

New generation of NTL Products

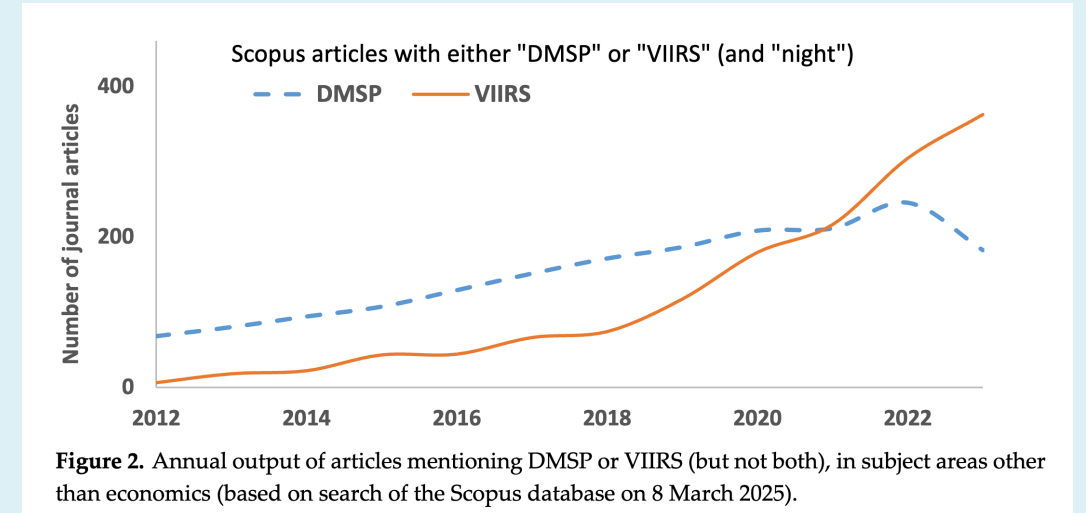
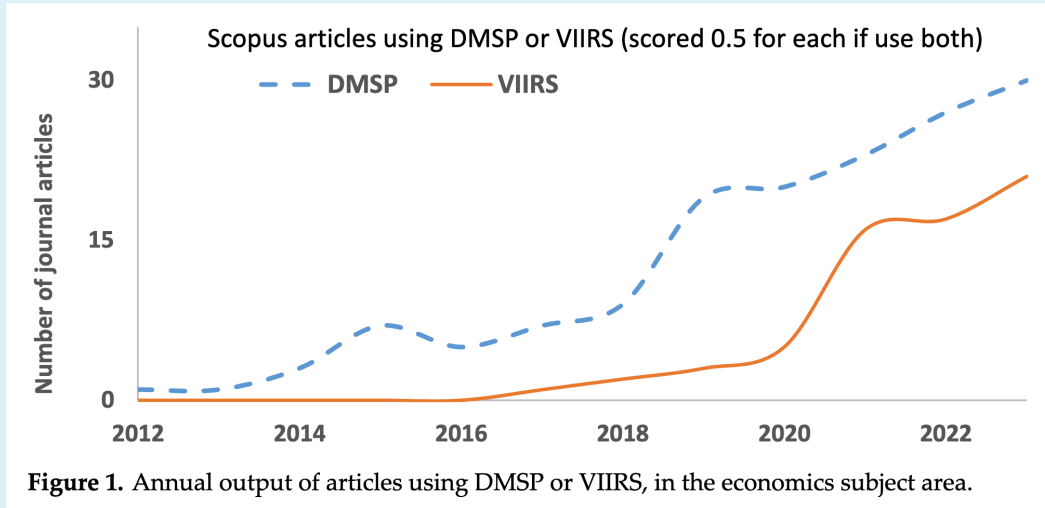
- VIIRS/DNB (Visible Infrared Imaging Radiometer Suite/Day Night Band)
- Operated by the U.S. National Oceanic and Atmospheric Administration (NOAA) and NASA
- from 2012 to present and daily, monthly, and annual products, high resolution

Attribute	DMSP/OLS	VIIRS DNB
Orbital Details	Sun-synchronous, ~850 km	Sun-synchronous, 824 km
Nighttime Overpass Time	~1930 UTC	~0130 UTC
Swath Width	3000 km	3000 km
Spectral Passband Bandwidth	Panchromatic 500–900 nm	Panchromatic 500–900 nm
Spectral Passband Center	~600 nm	~700 nm
Horizontal Spatial Resolution	5 km (Nadir)/~7 km (Edge)	<0.770 km (Scan) <0.750 km (Track)
Minimum Detectable Signal	$4 \times 10^{-5} \text{ W} \cdot \text{m}^{-2} \cdot \text{sr}^{-1}$	$3 \times 10^{-5} \text{ W} \cdot \text{m}^{-2} \cdot \text{sr}^{-1}$
Noise Floor	$\sim 5 \times 10^{-6} \text{ W} \cdot \text{m}^{-2} \cdot \text{sr}^{-1}$	$\sim 5 \times 10^{-7} \text{ W} \cdot \text{m}^{-2} \cdot \text{sr}^{-1}$ (Nadir)
Radiometric Quantization	6 bit	13–14 bit

DMSP/OLS vs VIIRS/DNB in resolution



DMSP/OLS vs VIIRS/DNB using in Economics



- Economics applications of NTL data are more likely to use both DMSP/OLS and VIIRS/DNB.
- Other fields of study are more likely to use VIIRS/DNB in recent years.

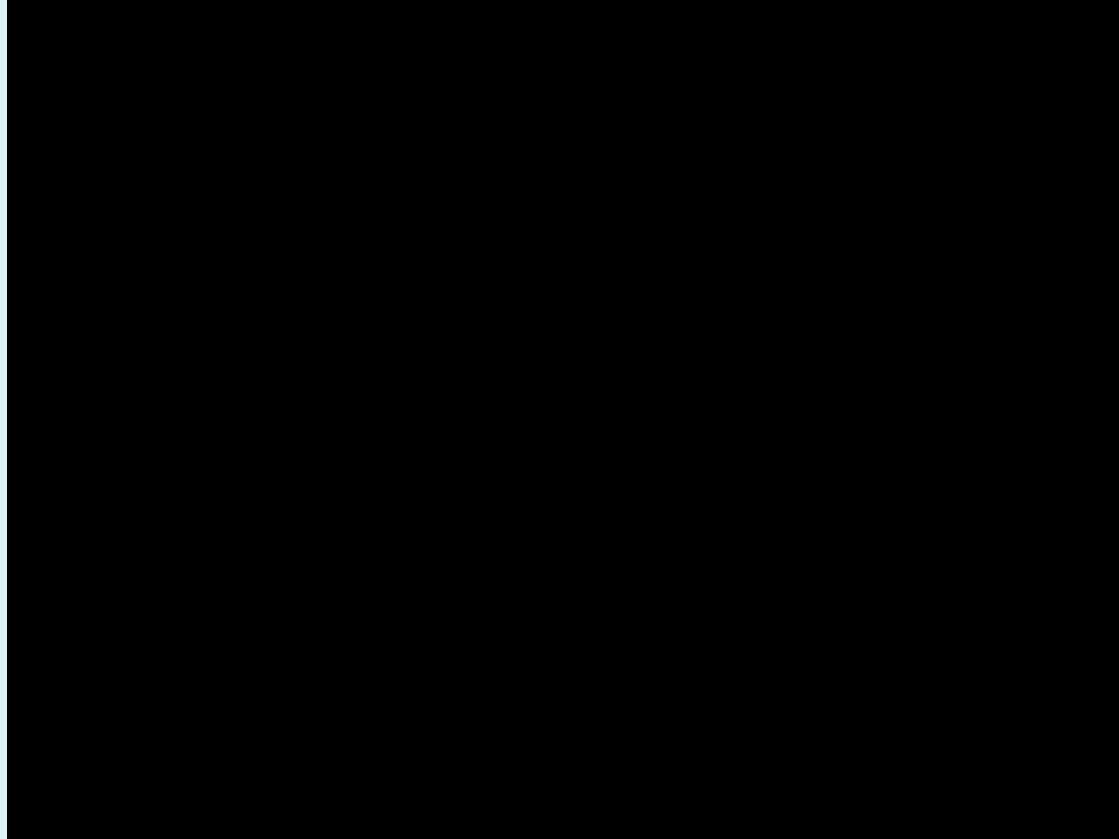
The Products of VIIRS/DNB nightlights data

Colorado School of Mines(CSM) Product and NASA's Black Marble (VNP46) Data Product

	NOAA VIIRS V1	NASA Black Marble V1
Frequency	Monthly and annual composites	Daily, monthly, and annual composites
Moonlight	Uses moon-free nights	Uses all nights and corrects lunar reflectance effects
Snow	Uncorrected for snow reflectance	Corrects for snow BRDF effects
Stray light	Filtered	Filtered
Ephemeral Lights	Filtered (for Annual Products)	Not filtered
Atmospheric effects	Not corrected	Corrected
Clouds	Cloud-free	Cloud-free

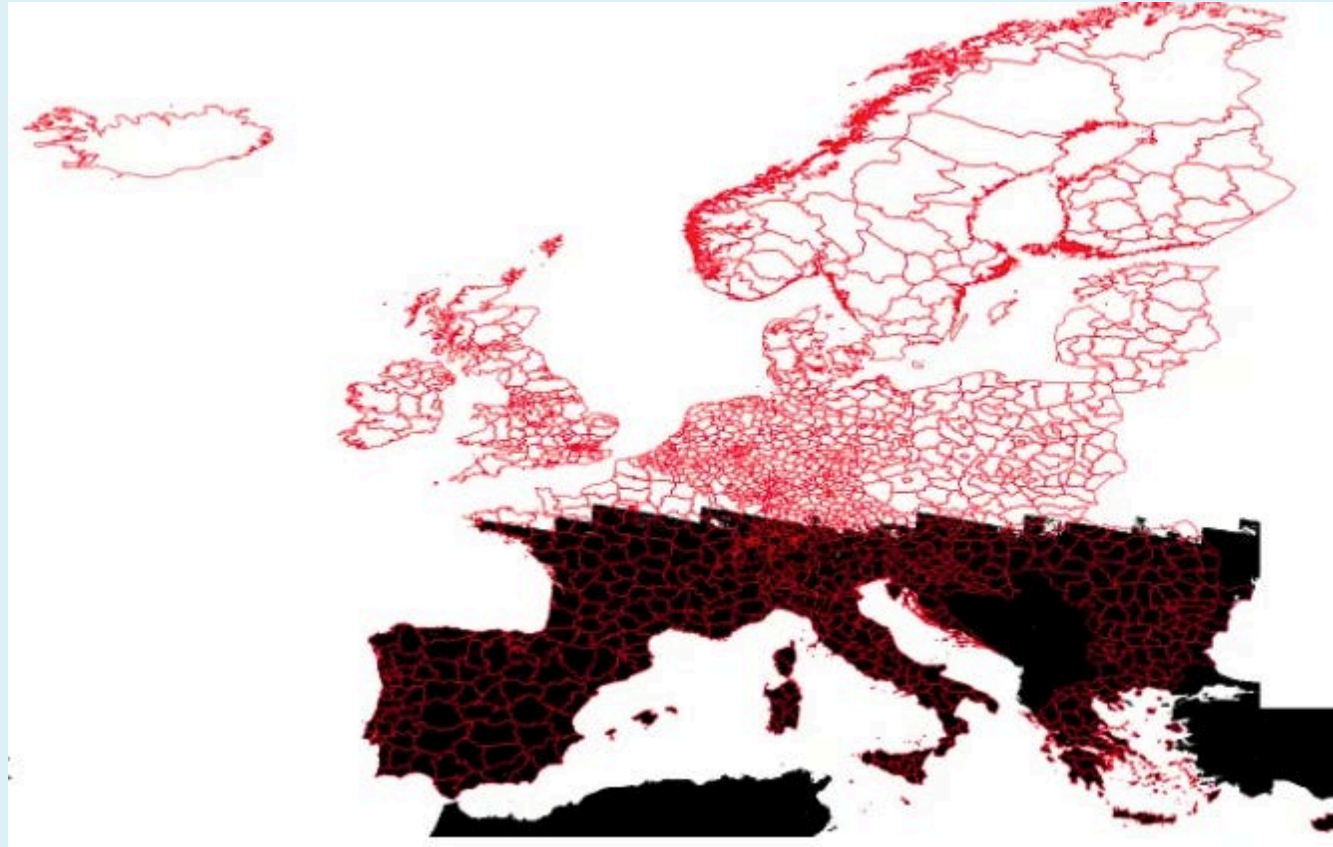
For more information about the algorithm, refer to Román, M.O., Wang, Z., Sun, Q., Kalb, V., Miller, S.D., Molthan, A., Schultz, L., Bell, J., Stokes, E.C., Pandey, B. and Seto, K.C., 2018. NASA's Black Marble nighttime lights product suite. *Remote Sensing of Environment*, 210, pp.113-143.

The night lights data are full of noise



- Surface reflectance of the earth's surface(snow, ice, etc.)
- Atmospheric reflection(because of air pollution or cloud)
- Solar,moon and stars reflection(because of the sunlight or moonlight)

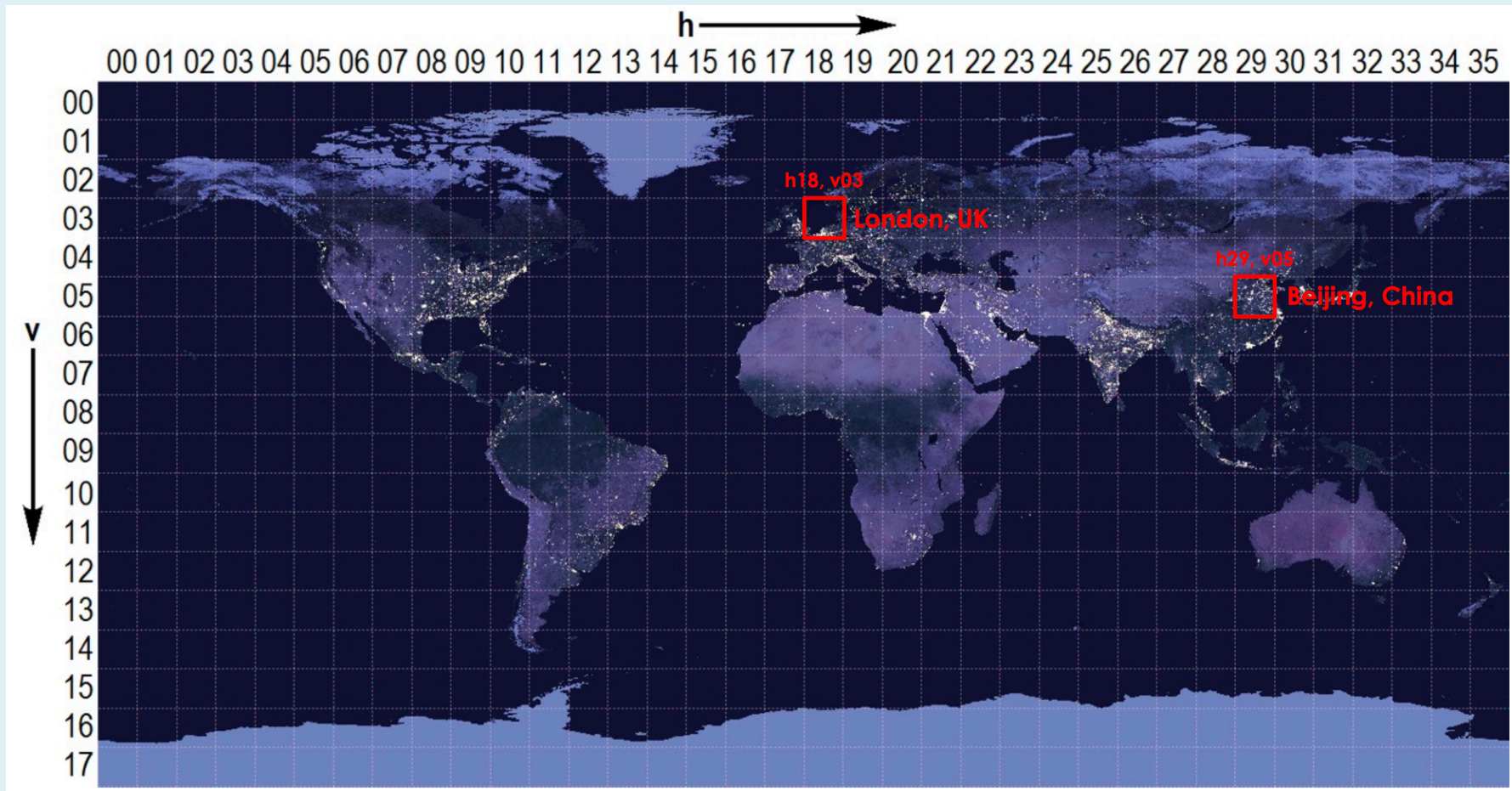
Stray Light Problem of VIIRS/DNB nightlights data



- Because Black Marble (VNP46) drops all pixels contaminated by stray light, it has many missing values for northern countries from June to August each year.

NASA's Black Marble (VNP46) grid structure

- The Suomi-NPP VIIRS linear latitude/longitude (or geographic) grid consists of 460 non-overlapping land tiles which cover approximately $10^\circ \times 10^\circ$ regions.



NASA's Black Marble (VNP46) Data Product

- Four different products of Black Marble Data are available:
 - VNP46A1, VNP46A2, VNP46A3, VNP46A4

Product ID	Distribution Service	Geographic Coverage	Temporal Coverage	Spatial Resolution	Temporal Resolution	Availability	Production Status	Product Availability
VNP46A1	Free/Open Access	Global	2012-YTD	15 arc second	Daily	< 1 day	In operations (2012-YTD)	Now - NASA LAADS Archive and Google EarthEngine
VNP46A2	Free/Open Access	Global	2012-YTD	15 arc second	Daily	< 1 day	In operation (2012-YTD)	Now - NASA LAADS Archive and Google EarthEngine
VNP46A3	Free/Open Access	Global	2012-YTD	15 arc second	Monthly	1 week	In operation (2012-YTD)	Now - NASA LAADS Archive
VNP46A4	Free/Open Access	Global	2012-YTD	15 arc second	Annual	1 week	In operation (2012-YTD)	Now - NASA LAADS Archive
Black Marble HD	Processing On-Demand*	Local-to-Regional	2012-YTD	≤ 30 m	Daily/Monthly/Annual	< 1 week	Processing On-Demand*	Processing On-Demand*
Black Marble NRT	Free/Open Access	Global	2012-YTD	15 arc second	Daily	< 1 day	In operation (2012-YTD)	Lance - Daily AND Hourly Gridded

Scientific Data Sets (SDS HDF Layers)	Units	Bit Types	Fill Value	Valid Range
DNB_BRDF-Corrected_NTL	$\text{nWatts} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$	16-bit unsigned integer	65,535	0 – 65,534
Gap_Filled_DNB_BRDF-Corrected_NTL	$\text{nWatts} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$	16-bit unsigned integer	65,535	0 – 65,534
DNB_Lunar_Irradiance	$\text{nWatts} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$	16-bit unsigned integer	65,535	0 – 65,534
Mandatory_Quality_Flag	Class Flag	8-bit unsigned integer	255	0 – 3
Latest_High_Quality_Retrieval	Number of days	8-bit unsigned integer	255	0 – 254
Snow_Flag	Class Flag	8-bit unsigned integer	255	0 – 1
QF_Cloud_Mask	Class Flag	16-bit unsigned integer	65,535	0 – 65,534

Data Processing and Analysis of NTL Data

Data Processing

Download NTL Data

1. Download NTL Data from NASA server
 - NASA's Level-1 Land and Atmosphere Archive and Distribution System Distributed Active Archive Center (LAADS-DAAC)
2. Using Packages like `blackmarbleR` to download NTL Data
3. by Google Earth Engine

Homework

- Register an account in NASA Earthdata [here](#)
- Read the guide of how to download NTL Data from NASA Earthdata
- Try to download NTL Data of China in 2020 from NASA Earthdata